# 1.35 2D Matrices as Linear Transformations

Worked Solutions

**1 a)**

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & a & a+b & b \\ 0 & 2 & 2+d & d \end{pmatrix} = \begin{pmatrix} 0 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \end{pmatrix}$$

$$\therefore a = 1,\ b = 2,\ c = 2,\ d = 1$$

So $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

**1 b)**

$$\begin{pmatrix} -1 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 0 & 3 & 4 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & a & a+b & b \\ 0 & 3 & 3+d & d \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 0 & 3 & 4 & 1 \end{pmatrix}$$

$$\therefore a = -1,\ b = 1,\ c = 3,\ d = 1$$

So $\begin{pmatrix} -1 & 1 \\ 3 & 1 \end{pmatrix}$.

**1 c)**

$$\begin{pmatrix} -1 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -3 & -2 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & a & a+b & b \\ 0 & 1 & 1+d & d \end{pmatrix} = \begin{pmatrix} 0 & -1 & -3 & -2 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

$$\therefore a = -1,\ b = -2,\ c = 1,\ d = -1$$

So $\begin{pmatrix} -1 & -2 \\ 1 & -1 \end{pmatrix}$.

**2 a)**

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 2 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} b & 2b & a+2b \\ d & 2d & c+2d \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 2 & 1 \end{pmatrix}$$

$$b = 1,\ d = 1$$

$$a + 2 \times 1 = 4 \Rightarrow a = 2$$

$$c + 2 \times 1 = 1 \Rightarrow c = -1$$

So $\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$.

**2 b)**

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 2 & 4 \\ 2 & 3 & 0 \end{pmatrix} = \begin{pmatrix} -2 & -2 & 2 \\ 0 & -1 & -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 2b & 2a+3b & 4a \\ 2d & 2c+3d & 4c \end{pmatrix} = \begin{pmatrix} -2 & -2 & 2 \\ 0 & -1 & -2 \end{pmatrix}$$

$$4a = 2 \Rightarrow a = \frac{1}{2} \qquad 2b = -2 \Rightarrow b = -1 \qquad 4c = -2 \Rightarrow c = -\frac{1}{2} \qquad 2d = 0 \Rightarrow d = 0$$

So $\begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{1}{2} & 0 \end{pmatrix}$.

**2 c)**

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 2 & -2 & 2 \\ -2 & 4 & 2 \end{pmatrix} = \begin{pmatrix} -3 & 3 & -3 \\ 3 & -6 & -3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2a-2b & -2a+4b & 2a+2b \\ 2c-2d & -2c+4d & 2c+2d \end{pmatrix} = \begin{pmatrix} -3 & 3 & -3 \\ 3 & -6 & -3 \end{pmatrix}$$

$$2a - 2b = -3 \quad (1) \qquad -2a + 4b = 3 \quad (2)$$

$$\therefore a = -\frac{3}{2} \text{ and } b = 0$$

$$2c - 2d = 3 \quad (3) \qquad -2c + 4d = -6 \quad (4)$$

$$\therefore c = 0 \text{ and } d = -\frac{3}{2}$$

So $\begin{pmatrix} -\frac{3}{2} & 0 \\ 0 & -\frac{3}{2} \end{pmatrix}$.

**3 a)**

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

So (2, 3).

**3 b)**

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$$

So (4, 9).

**3 c)**

$$\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 11 \\ 3 \end{pmatrix}$$

So (11, 3).

**3 d)**

$$\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 13 \end{pmatrix}$$

So (2, 13).

**3 e)**

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$

So (5, 5).

**3 f)**

$$\begin{pmatrix} -1 & 2 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \end{pmatrix}$$

So (4, -6).

**4 a)**

This is the identity matrix. All points get mapped to themselves.

**4 b)**

This is the zero matrix. All points get mapped to the origin, $(0, 0)$.

**5 a)**

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ -5 & -1 \end{pmatrix} = \begin{pmatrix} -16 & 2 \\ -16 & 2 \end{pmatrix}$$

So $A'(-16, -16)$ and $B'(2, 2)$.

**5 b)**

$$\begin{pmatrix} -3 & 1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ -5 & -1 \end{pmatrix} = \begin{pmatrix} 4 & -7 \\ 2 & -10 \end{pmatrix}$$

So $A'(4, 2)$ and $B'(-7, -10)$.

**5 c)**

$$\begin{pmatrix} 1.2 & 3.6 \\ -2.8 & -1.9 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ -5 & -1 \end{pmatrix} = \begin{pmatrix} -21.6 & -1.2 \\ 17.9 & -3.7 \end{pmatrix}$$

So $A'(-21.6, 17.9)$ and $B'(-1.2, -3.7)$.

**5 d)**

$$\begin{pmatrix} 0 & 0 \\ 0 & 7.7 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ -5 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -38.5 & -7.7 \end{pmatrix}$$

So $A'(0, -38.5)$ and $B'(0, -7.7)$.

**6 a)**

$PQ = \sqrt{(4-2)^2 + (-3-1)^2} = \sqrt{20} = 2\sqrt{5}$

**6 b)**

$$\begin{pmatrix} 5 & -3 \\ 4 & -8 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} -1 & -27 \\ -12 & -44 \end{pmatrix}$$

So $P'(-1, -12)$ and $Q'(-27, -44)$.

$$P'Q' = \sqrt{(-44-(-12))^2 + (-27-(-1))^2} = \sqrt{1700} = 10\sqrt{17}$$

**7)**

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 4 & -2 \\ -1 & -3 \end{pmatrix} = \begin{pmatrix} 7 & -7 \\ 1 & -11 \end{pmatrix} \Rightarrow \begin{pmatrix} 4a-b & -2a-3b \\ 4c-d & -2c-3d \end{pmatrix} = \begin{pmatrix} 7 & -7 \\ 1 & -11 \end{pmatrix}$$

$4a - b = 7 \quad (1) \qquad -2a - 3b = -7 \quad (2) \qquad \therefore a = 2 \text{ and } b = 1$

$4c - d = 1 \quad (3) \qquad -2c - 3d = -11 \quad (4) \qquad \therefore c = 1 \text{ and } d = 3$

So $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$.

**8)**

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 5 & 2 \\ 6 & -1 \end{pmatrix} = \begin{pmatrix} -37 & -8 \\ 50 & 3 \end{pmatrix} \Rightarrow \begin{pmatrix} 5a+6b & 2a-b \\ 5c+6d & 2c-d \end{pmatrix} = \begin{pmatrix} -37 & -8 \\ 50 & 3 \end{pmatrix}$$

$5a + 6b = -37 \quad (1) \qquad 2a - b = -8 \quad (2) \qquad \therefore a = -5 \text{ and } b = -2$

$5c + 6d = 50 \quad (3) \qquad 2c - d = 3 \quad (4) \qquad \therefore c = 4 \text{ and } d = 5$

So $\begin{pmatrix} -5 & -2 \\ 4 & 5 \end{pmatrix}$.

**9)**

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} a+2b & a+b \\ c+2d & c+d \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$a + 2b = 1 \quad (1) \qquad a + b = 1 \quad (2) \qquad \therefore a = 1 \text{ and } b = 0$

$c + 2d = 1 \quad (3) \qquad c + d = 2 \quad (4) \qquad \therefore c = 3 \text{ and } d = -1$

So $\begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix}$.

**10)**

$$\begin{pmatrix} 6 & 8 \\ -2 & 9 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 46 \\ 78 \end{pmatrix} \Rightarrow \begin{pmatrix} 6a+8b \\ -2a+9b \end{pmatrix} = \begin{pmatrix} 46 \\ 78 \end{pmatrix}$$

$$6a+8b=46 \quad (1) \qquad -2a+9b=78 \quad (2)$$

Solve (1) and (2) to give $a=-3$ and $b=8$.

**11)**

$$\begin{pmatrix} 4 & 10 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 8 & 76 \\ -5 & 13 \end{pmatrix} \Rightarrow \begin{pmatrix} 4a+10b & 4c+10d \\ 3a+2b & 3c+2d \end{pmatrix} = \begin{pmatrix} 8 & 76 \\ -5 & 13 \end{pmatrix}$$

$$4a+10b=8 \quad (1) \qquad 3a+2b=-5 \quad (2) \qquad \therefore a=-3 \text{ and } b=2$$

$$4c+10d=76 \quad (3) \qquad 3c+2d=13 \quad (4) \qquad \therefore c=-1 \text{ and } d=8$$

So $\begin{pmatrix} -3 & -1 \\ 2 & 8 \end{pmatrix}$.

$a=-3$, $b=2$, $c=-1$, $d=8$.

**12)**

$$\begin{pmatrix} 3 & k \\ -4 & 2k \end{pmatrix}\begin{pmatrix} a \\ 16 \end{pmatrix} = \begin{pmatrix} -23 \\ -236 \end{pmatrix} \Rightarrow \begin{pmatrix} 3a+16k \\ -4a+32k \end{pmatrix} = \begin{pmatrix} -23 \\ -236 \end{pmatrix}$$

$$3a+16k=-23 \quad (1) \qquad -4a+32k=-236 \quad (2)$$

Solve (1) and (2) to give $k=-5$ and $a=19$.

**13)**

$$\begin{pmatrix} 4k & -5k \\ 8k & 3k \end{pmatrix}\begin{pmatrix} k \\ 8 \end{pmatrix} = \begin{pmatrix} -52k \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 4k^2-40k \\ 8k^2+24k \end{pmatrix} = \begin{pmatrix} -52k \\ 0 \end{pmatrix}$$

$$4k^2-40k=-52k \Rightarrow 4k^2+12k=0 \Rightarrow k^2+3k=0$$

$$\Rightarrow k(k+3)=0 \Rightarrow k=0 \text{ or } k=-3$$

$$8k^2+24k=0 \Rightarrow k^2+3k=0 \Rightarrow k(k+3)=0 \Rightarrow k=0 \text{ or } k=-3$$

As $k \neq 0$, we must have that $k=-3$.

**14)**

$$\begin{pmatrix} k+3 & k-5 \\ k-7 & k+1 \end{pmatrix}\begin{pmatrix} 3k \\ k-6 \end{pmatrix} = \begin{pmatrix} 44k+8 \\ 17k+5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} (k+3)\times 3k+(k-5)\times(k-6) \\ (k-7)\times 3k+(k+1)\times(k-6) \end{pmatrix} = \begin{pmatrix} 44k+8 \\ 17k+5 \end{pmatrix}$$

$$3k^2+9k+k^2-11k+30=44k+8 \Rightarrow 4k^2-46k+22=0$$

$$\Rightarrow 2k^2-23k+11=0 \Rightarrow (2k-1)(k-11)=0 \Rightarrow k=\frac{1}{2} \text{ or } k=11$$

$$3k^2-21k+k^2-5k-6=17k+5 \Rightarrow 4k^2-43k-11=0$$

$$\Rightarrow (4k+1)(k-11)=0 \Rightarrow k=-\frac{1}{4} \text{ or } k=11$$

$$\therefore k=11$$

**15)**

$$\begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} 2k+3 \\ k-1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} k \\ 1 \end{pmatrix} = \begin{pmatrix} 3k+2 \\ 1-k \end{pmatrix}$$

So $P'(2k+3, k-1)$ and $Q'(3k+2, 1-k)$.

$$P'Q' = 2\sqrt{5} \Rightarrow ((3k+2)-(2k+3))^2 + ((1-k)-(k-1))^2 = (2\sqrt{5})^2$$

$$\Rightarrow (k-1)^2 + (2-2k)^2 = 20$$

$$\Rightarrow k^2 - 2k + 1 + 4 - 8k + 4k^2 = 20$$

$$\Rightarrow 5k^2 - 10k - 15 = 0$$

$$\Rightarrow k^2 - 2k - 3 = 0$$

$$(k-3)(k+1) = 0$$

$$\Rightarrow k = -1 \text{ or } k = 3$$

**16)**

Choose two points, $(1, 0)$ and $(0, 1)$ for example, and find their images:

$$\begin{pmatrix} 8 & -6 \\ 4 & -3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} 8 & -6 \\ 4 & -3 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -3 \end{pmatrix}$$

The equation of the line passing through $(8, 4)$ and $(-6, -3)$ is:

$$y - 4 = \frac{-3-4}{-6-8}(x-8) \Rightarrow y - 4 = \frac{1}{2}(x-8) \Rightarrow y = \frac{1}{2}x$$

**17)**

$$\begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ x \end{pmatrix} = \begin{pmatrix} X \\ aX \end{pmatrix} \Rightarrow \begin{pmatrix} x-3x \\ 2x+x \end{pmatrix} = \begin{pmatrix} X \\ aX \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2x \\ 3x \end{pmatrix} = \begin{pmatrix} X \\ aX \end{pmatrix}$$

$$X = -2x \qquad aX = 3x$$

$$\therefore a(-2x) = 3x \Rightarrow a = -\frac{3}{2}$$

The image of the line is $y = -\frac{3}{2}x$, so $a = -\frac{3}{2}$.

Alternative Method:

Find the image of two points that lie on the line $y = x$:

$$\begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix}1 & -3\\ 2 & 1\end{pmatrix}\begin{pmatrix}2\\ 2\end{pmatrix}=\begin{pmatrix}-4\\ 6\end{pmatrix}$$

Find the equation of the line that passes through $(-2,3)$ and $(-4,6)$:

$$m=\frac{6-3}{-4-(-2)}=\frac{3}{-2}=-\frac{3}{2}$$

$$\therefore y-3=-\frac{3}{2}(x+2)\Rightarrow y=-\frac{3}{2}x,\ \text{so } a=-\frac{3}{2}$$

**18)**

$$\begin{pmatrix}-2 & -5\\ 8 & -3\end{pmatrix}\begin{pmatrix}x\\ x-2\end{pmatrix}=\begin{pmatrix}X\\ aX+b\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}-2x-5(x-2)\\ 8x-3(x-2)\end{pmatrix}=\begin{pmatrix}X\\ aX+b\end{pmatrix}\Rightarrow\begin{pmatrix}10-7x\\ 6+5x\end{pmatrix}=\begin{pmatrix}X\\ aX+b\end{pmatrix}$$

$$X=10-7x\qquad aX+b=6+5x$$

$$\therefore a(10-7x)+b=6+5x\Rightarrow 10a+b-7ax=6+5x$$

$$10a+b=6\quad(1)\qquad -7a=5\quad(2)$$

Solve (1) and (2) to get $a=-\frac{5}{7}$ and $b=\frac{92}{7}$.

The image of the line is $y=-\frac{5}{7}x+\frac{92}{7}$.

So $a=-\frac{5}{7}$ and $b=\frac{92}{7}$.

Alternative Method:

Find the image of two points that lie on the line $y=x-2$:

$$\begin{pmatrix}-2 & -5\\ 8 & -3\end{pmatrix}\begin{pmatrix}2\\ 0\end{pmatrix}=\begin{pmatrix}-4\\ 16\end{pmatrix}$$

$$\begin{pmatrix}-2 & -5\\ 8 & -3\end{pmatrix}\begin{pmatrix}0\\ -2\end{pmatrix}=\begin{pmatrix}10\\ 6\end{pmatrix}$$

Find the equation of the line that passes through $(-4,16)$ and $(10,6)$:

$$m=\frac{16-6}{-4-10}=\frac{10}{-14}=-\frac{5}{7}$$

$$\therefore y-16=-\frac{5}{7}(x+4)\Rightarrow y=-\frac{5}{7}x+\frac{92}{7}$$

So $a=-\frac{5}{7}$ and $b=\frac{92}{7}$.

**19)**

$$\begin{pmatrix}a & 3\\ 2 & b\end{pmatrix}\begin{pmatrix}x\\ 2x+1\end{pmatrix}=\begin{pmatrix}X\\ 3X+2\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}ax+3(2x+1)\\ 2x+b(2x+1)\end{pmatrix}=\begin{pmatrix}X\\ 3X+2\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}(a+6)x+3\\ (2+2b)x+b\end{pmatrix}=\begin{pmatrix}X\\ 3X+2\end{pmatrix}$$

$$X = (a+6)x + 3 \qquad 3X + 2 = (2+2b)x + b$$
$$\therefore 3((a+6)x+3) + 2 = (2+2b)x + b \Rightarrow (3a+18)x + 11 = (2+2b)x + b$$
$$3a + 18 = 2 + 2b \quad (1) \qquad 11 = b \quad (2)$$

Substitute (2) into (1):

$$3a + 18 = 2 + 2(11) \Rightarrow a = 2$$
$$\therefore a = 2 \text{ and } b = 11$$

Alternative Method:

Find the image of two points that lie on the line $y = 2x + 1$:

$$\begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ b \end{pmatrix}$$
$$\begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} a+9 \\ 2+3b \end{pmatrix}$$

Substitute $(3, b)$ and $(a+9, 2+3b)$ into $y = 3x + 2$:

$$(3, b) \Rightarrow b = 3(3) + 2 \Rightarrow b = 11$$
$$(a+9, 2+3b) \Rightarrow 2 + 3b = 3(a+9) + 2$$
$$\Rightarrow 2 + 3(11) = 3a + 29 \Rightarrow a = 2$$
$$\therefore a = 2 \text{ and } b = 11$$

**20)**

$(4, 0)$ is the centre of the circle.

$(2, 2)$ lies on the circle.

$$\begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ 12 & 6 \end{pmatrix}$$

So $C'$ has centre $(0, 12)$ and passes through $(-6, 6)$.

$$r' = \sqrt{(-6-0)^2 + (6-12)^2} = \sqrt{72}$$
$$\therefore C' \text{ has equation } x^2 + (y-12)^2 = 72$$

**21)**

$$\begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ x^2 + 2x \end{pmatrix} = \begin{pmatrix} X \\ aX^2 + bX + c \end{pmatrix}$$
$$\Rightarrow \begin{pmatrix} -2x \\ -3x^2 - 6x \end{pmatrix} = \begin{pmatrix} X \\ aX^2 + bX + c \end{pmatrix}$$
$$X = -2x$$
$$aX^2 + bX + c = -3x^2 - 6x$$
$$\therefore a(-2x)^2 + b(-2x) + c = -3x^2 - 6x$$
$$\Rightarrow 4ax^2 - 2bx + c = -3x^2 - 6x$$
$$4a = -3 \Rightarrow a = -\frac{3}{4} \qquad -2b = -6 \Rightarrow b = 3 \qquad c = 0$$

The image of the curve is $y = -\frac{3}{4}x^2 + 3x$.

So $a = -\frac{3}{4}$, $b = 3$ and $c = 0$.

**22)**

$$\begin{pmatrix} -2 & -2 \\ 6 & -6 \end{pmatrix} \begin{pmatrix} x \\ \frac{1}{x} \end{pmatrix} = \begin{pmatrix} X \\ \sqrt{\frac{a}{b}X^2 - \frac{1}{b}} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2x - \frac{2}{x} \\ 6x - \frac{6}{x} \end{pmatrix} = \begin{pmatrix} X \\ \sqrt{\frac{a}{b}X^2 - \frac{1}{b}} \end{pmatrix}$$

$$X = -2x - \frac{2}{x}$$

$$\sqrt{\frac{a}{b}X^2 - \frac{1}{b}} = 6x - \frac{6}{x}$$

$$\therefore \sqrt{\frac{a}{b}(-2x - \frac{2}{x})^2 - \frac{1}{b}} = 6x - \frac{6}{x}$$

$$\Rightarrow \frac{a}{b}(-2x - \frac{2}{x})^2 - \frac{1}{b} = (6x - \frac{6}{x})^2$$

$$\Rightarrow a(4x^2 + 8 + \frac{4}{x^2}) - 1 = b(36x^2 - 72 + \frac{36}{x^2})$$

$$\Rightarrow 4ax^2 + \frac{4a}{x^2} + 8a - 1 = 36bx^2 + \frac{36b}{x^2} - 72b$$

$$4a = 36b \Rightarrow a - 9b = 0 \quad (1) \qquad 8a - 1 = -72b \Rightarrow 8a + 72b = 1 \quad (2)$$

Solve (1) and (2) to get $a = \frac{1}{16}$ and $b = \frac{1}{144}$.

The image of the curve is $\frac{1}{16}x^2 - \frac{1}{144}y^2 = 1$.

So $a = \frac{1}{16}$ and $b = \frac{1}{144}$.